

A Geometric-Null Binomial Framework for Harmonic Longitude Proximity Testing in Monument Distributions

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Highlights

- Geometric-null binomial is the test matched to harmonic-proximity hypotheses.
- Rayleigh concentration and binomial proximity enrichment can dissociate.
- Dissociation demonstrated on the UNESCO dome/stupa corpus ($N = 90$).
- Worked example: Wiki+OSM Fisher $\chi^2(4) = 28.74$, $p < 0.001$, T3, excluding domes.
- Anchor-sweep diagnostic on UNESCO corpus: max-statistic $p = 0.0753$ (marginal).

Abstract

This paper addresses a mismatch between test and hypothesis in spatial periodicity analysis. The Rayleigh statistic measures mean phase concentration at period T , whereas a repeated-line grid hypothesis predicts thresholded proximity to any harmonic longitude. These two summaries can dissociate within the same corpus. The methods contribution is the geometric-null binomial framework: (i) define phases $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$; (ii) compute harmonic deviation $\delta_i = \min(r_i, T - r_i)$; (iii) test the count of sites with $\delta_i \leq d$ against the geometric null $p_0 = 2d/T$; (iv) combine across independently assembled corpora via Fisher's method; and

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(v) apply an anchor sweep correction based on a max statistic when the anchor requires validation. A period sweep of the UNESCO dome/stupa corpus ($N = 90$) demonstrates the dissociation between Rayleigh concentration and proximity enrichment. Rayleigh is significant at three periods that lack proximity enrichment, while the 3° period shows proximity enrichment with non-significant Rayleigh. Applied to three hemispherical-monument corpora at the 3° period and ~35.25°E Levantine corridor phase, the worked example finds consistent corpus-level proximity enrichment. The result is invariant within the band, so any longitude on the same grid within $\pm 0.06^\circ$ of this phase, including Jerusalem and Mount Gerizim at 35.269°E, produces identical proximity counts. The two-corpus combination that excludes the dome corpus (Wikidata + OSM) gives $\chi^2(4) = 28.74$, $p < 0.001$ at T3 (threshold $\pm 0.10 \times T$). A max-statistic permutation test over 36,000 trial longitudes applied to the UNESCO corpus yields $p = 0.0753$ (marginal), indicating that the Levantine corridor phase is not a strongly exceptional anchor within the UNESCO corpus once the 36,000-anchor search is accounted for. This test is a diagnostic on the UNESCO corpus only; no equivalent anchor sweep was run on Wikidata or OSM, whose results are simply frame-conditional on the anchor identified in the dome sweep. Following [Silva \(2020\)](#), all results are treated as statistical descriptions of spatial structure; questions of mechanism and historical transmission require independent evidence.

Keywords: spatial statistics, circular statistics, harmonic proximity, binomial geometric null, monument distribution, archaeoastronomy

1 Introduction

Do monuments cluster near specific longitude intervals? The question recurs in the archaeoastronomical and archaeogeographic literature, usually motivated by observational coincidences involving sacred sites, trade corridors, or cosmographic traditions. This paper focuses on *hemispherical monuments*, defined here as structures whose principal form is a rounded or mounded mass, including Buddhist stupas, corbelled tholos tombs, and

domed shrines, as a corpus for testing longitude-grid hypotheses. This broader question has also been criticized on grounds of investigator-assembled site lists, insufficient correction for multiple comparisons, and lack of independent replication (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). These concerns motivate a stricter separation between statistical detection and archaeological explanation. Alongside these critiques, this paper raises the complementary issue of matching tests to hypotheses. The standard circular-statistical tool used in this literature measures a different quantity from the one predicted by a repeated-line grid hypothesis.

The tool in question is the Rayleigh statistic R . In its circular-statistical form, Rayleigh’s mean-resultant test is a standard test of departure from uniformity toward a single preferred direction on the circle (Rayleigh, 1919; Fisher, 1993; Mardia & Jupp, 2000), most commonly encountered in archaeology and skyscape work in orientation and azimuth analysis (Magli, 2016; Silva, 2020). Here it becomes relevant because longitudes can be folded modulo a candidate period T : R then measures whether the folded phases concentrate toward one direction.

This is a useful diagnostic, but only a partial answer to the repeated-line question. A grid-placement hypothesis predicts that sites fall near longitudes $\lambda_0, \lambda_0 + T, \lambda_0 + 2T, \dots$. Under modulo- T phase mapping these harmonics collapse to a common phase, so an ideal grid-aligned corpus would produce strong Rayleigh concentration. In real geographically clustered corpora, however, Rayleigh concentration and thresholded harmonic proximity can dissociate: a dominant off-harmonic cluster can produce significant Rayleigh concentration, while a corpus with many near-harmonic sites plus substantial off-harmonic mass can produce binomial enrichment without significant Rayleigh R . A non-significant Rayleigh result therefore does not rule out harmonic proximity, and a significant Rayleigh result does not by itself establish proximity to repeated longitude lines. The two statistics are complementary; they answer related but distinct questions.

The directly matched test for “sites cluster near harmonics” is straightforward. Under a uniform longitude model, the expected fraction of longitudes falling within $\pm d^\circ$ of any harmonic of period T is $p_0 = 2d/T$. This is the *geometric null*. In clustered corpora it

remains a useful geometric benchmark, but the resulting p -values are interpreted here as corpus-level proximity enrichment rather than counts of independent placement decisions.

This paper develops that matched test and uses the monument data as a worked methodological example. Under a fixed period, fixed corridor phase, fixed proximity thresholds, and fixed operational corpora, the geometric-null binomial test estimates whether more sites fall near harmonic longitudes than expected from the geometric coverage of the threshold bands. Rayleigh statistics are retained as complementary circular diagnostics, and the period sweep is used to show when phase concentration and thresholded harmonic proximity agree or dissociate. The empirical application illustrates corpus-level proximity enrichment at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Following [Silva \(2020\)](#), this is treated throughout as statistical description of spatial structure in an open-data worked example; historical, cultural, or geographic explanation is a separate interpretive task requiring independent evidence.

2 Rayleigh Concentration and Harmonic Proximity: Complementary but Distinct Statistics

2.1 Why the Two Statistics Can Dissociate

Let $\lambda_1, \dots, \lambda_N$ be site longitudes and T a candidate grid period with anchor λ_0 . Define phases by $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$, and the Rayleigh statistic by

$$R = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} \right|,$$

where $i = \sqrt{-1}$ (complex exponential; equivalently, $e^{i\theta} = \cos \theta + i \sin \theta$, so R is the length of the mean of the unit vectors $(\cos \theta_i, \sin \theta_i)$).

A key property of this transformation is that sites near *any* harmonic of T all map to the same phase $\theta \approx 0$ under the modulo- T reduction. This includes sites near λ_0 , $\lambda_0 + T$, $\lambda_0 + 2T$, and so on, because $(\lambda_0 + kT - \lambda_0) \bmod T = 0$ for any integer k . Under ideal conditions, in which all sites fall exactly on a harmonic, R would equal 1. The modulo-

T phase map therefore does not, by itself, produce multiple phase classes: harmonics collapse to one class, not many.

The Rayleigh test and the binomial proximity test can nevertheless dissociate because they aggregate cross-site information differently.

What Rayleigh measures. R is the length of the mean phasor. It is sensitive to whether a *single direction* on the unit circle predominates after averaging over the full corpus. In a geographically clustered corpus, a single dominant geographic cluster can produce large R at period T simply because that cluster's longitude maps to a concentrated phase at that particular T , regardless of whether the cluster is harmonically proximate. This is a generic property of clustered data on the unit circle; it can produce significant R at periods that have no interpretation as harmonic proximity.

What the binomial measures. The binomial proximity test counts the number of sites whose deviation from the nearest harmonic falls within a threshold d , written as $\#\{i : \delta_i \leq d\}$. With $r_i = ((\lambda_i - \lambda_0) \bmod T)$ and $\delta_i = \min(r_i, T - r_i)$, this count is independent of the mean phase direction. Sites far from harmonics (large δ_i) contribute zero to the binomial count regardless of whether they push R up or down. Conversely, a corpus can show significant binomial enrichment even when many off-harmonic sites keep the mean phasor small.

Complementarity. The two tests ask complementary questions. Rayleigh asks whether one phase direction dominates. The binomial asks whether more sites lie within a fixed reporting threshold of a harmonic than the uniform null predicts. In a real corpus these can agree or disagree, depending on the balance between harmonically proximate sites, off-harmonic sites, and the geographic clustering structure of the corpus. A non-significant Rayleigh result does not rule out harmonic proximity; a significant Rayleigh result does not establish harmonic proximity unless the phase concentration is verified to coincide with a harmonic. Relying on Rayleigh alone to evaluate a repeated-line hypothesis can therefore produce false positives at cluster-dominated periods and miss genuine proximity enrichment when near-harmonic sites coexist with substantial off-harmonic mass.

2.2 Empirical Demonstration: Period Sweep Dissociation

The following demonstration uses the UNESCO keyword-matched dome/stupa corpus described fully in Section 4. Table 1 illustrates the dissociation concretely ($N = 90$). The full sweep of 46 candidate periods (1.5° to 6.0° in 0.1° steps) is available in the supplementary audit; four representative periods are shown here to illustrate the qualitatively distinct outcomes. Both tests use the T3 threshold ($\pm 0.10 \times T$, geometric null $p_0 = 20\%$; tiers defined in Section 3.2); finer-threshold tiers (T2, T1) are reported in the primary results.

Table 1: Rayleigh concentration vs. T3 binomial proximity enrichment at four representative periods drawn from a full sweep of 46 candidate periods (1.5° to 6.0° in 0.1° steps; complete sweep tables in supplementary audit). These four periods span the full range of dissociation outcomes, including three periods with significant Rayleigh concentration but no proximity enrichment, and one period with proximity enrichment but non-significant Rayleigh concentration. UNESCO dome corpus ($N = 90$, corridor phase $\sim 35.25^\circ\text{E}$). T3 threshold = $0.10 \times T$; geometric null $p_0 = 20\%$. Sig symbols: $*p < 0.05$, $**p < 0.01$, ns = not significant.

Period	Rayleigh p	Sig	T3 hits	Expected	Binom. p	Sig
2.4°	0.0083	**	21	18.0	0.2503	ns
3.0°	0.1796	ns	29	18.0	0.0043	**
3.5°	0.0267	*	13	18.0	0.9312	ns
3.6°	0.0407	*	20	18.0	0.3383	ns

The dissociation is clear. Rayleigh is significant at 2.4° , 3.5° , and 3.6° , and non-significant at 3.0° . The binomial proximity test is significant only at 3.0° . A researcher relying on Rayleigh alone to evaluate the repeated-line hypothesis would flag periods that carry no proximity enrichment, and would overlook the period with the clearest proximity structure. Using the two tests together avoids this.

The pattern is consistent with the mechanism described above. At 2.4° , 3.5° , and 3.6° , one geographic cluster dominates the phase distribution, producing concentrated R without harmonic proximity enrichment. At 3.0° , many sites fall near the harmonic class, but substantial off-harmonic mass keeps the mean phasor small; the binomial test correctly registers the thresholded per-site proximity counts. Rayleigh and the binomial measure genuinely different aspects of the data; neither is wrong, but only the binomial

directly answers the question “do sites fall near harmonics?”

A broader candidate-period survey of the dome/stupa corpus (1.5° to 6.0° in 0.1° steps, 46 periods; four representative periods shown in Table 1) finds the dominant binomial proximity peak at 3° ($p = 0.0043$); selected nearby non-harmonic spacings (1.8° , 2.1° , 2.7° , 3.3°) are non-significant in the same sweep. This motivates 3° as the primary period for the cross-corpus analysis in Section 5.

The dissociation also appears in the full UNESCO cultural corpus ($N = 1,011$ sites with coordinates, including non-dome cultural sites). At $T = 3^\circ$, Rayleigh remains non-significant ($p = 0.60$), while binomial proximity enrichment is significant across all three tier thresholds (T3: $p = 0.025$; T2: $p = 0.001$; T1: $p = 0.010$). Because this corpus includes the dome sites that motivated the period, it does not constitute independent validation. It does show that the dissociation between Rayleigh and binomial results is not an artifact of the restricted dome vocabulary.

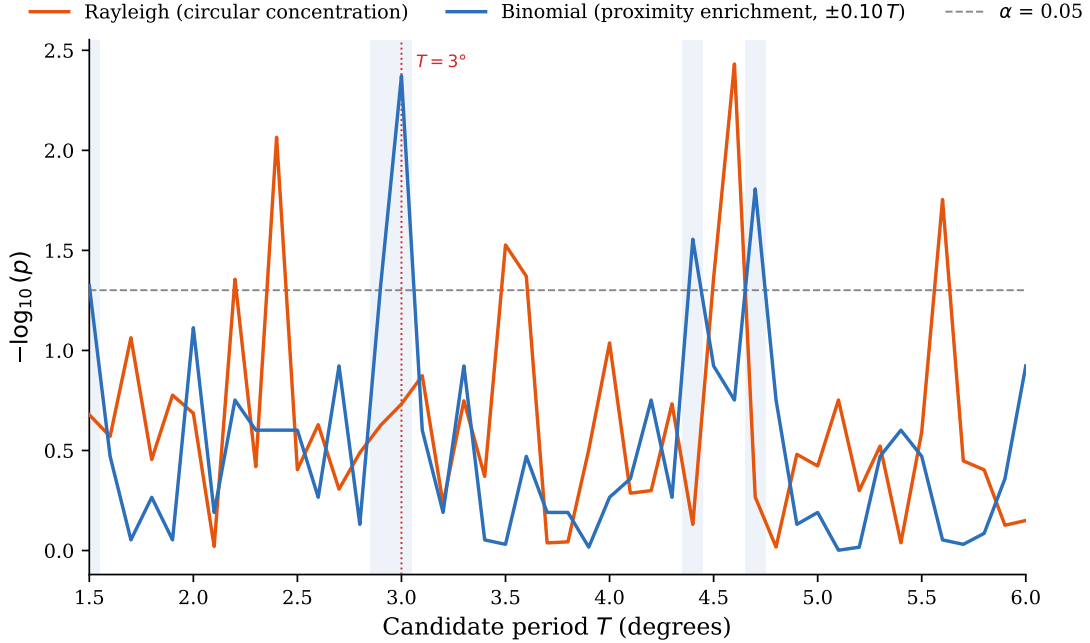


Figure 1: Period sweep of the UNESCO dome/stupa corpus ($N = 90$): $-\log_{10}(p)$ for the binomial proximity-enrichment test (blue; threshold $\pm 0.10 \times T$) and the permutation Rayleigh test (orange) across 46 candidate periods (1.5° to 6.0° in 0.1° steps); anchor $\approx 35.25^\circ\text{E}$. Dashed line marks $\alpha = 0.05$. The blue shaded column at $T = 3^\circ$ highlights the dominant proximity peak; Rayleigh is non-significant at this period, illustrating the dissociation between Rayleigh and binomial results.

A Monte Carlo spacing test (100,000 permutations, dome label shuffle) across 18

161 candidate spacings from 0.5° to 15° (full list in supplementary audit) finds the observed
 162 count of 4 significant spacings marginal ($p \approx 0.0979$). Three of the four significant
 163 spacings (6° , 8° , 12°) are multiples of 3° and their hits are fully accounted for by the 3°
 164 harmonic structure; the 3° spacing is the only one whose significance is not explained by
 165 overlap with another spacing’s nodes.

166 3 The Geometric-Null Binomial Framework

167 3.1 The Geometric Null

168 For a grid of period T anchored at λ_0 , define the anchor-relative remainder and nearest-
 169 harmonic deviation as

$$r_i = ((\lambda_i - \lambda_0) \bmod T), \quad \delta_i = \min(r_i, T - r_i).$$

170 A site falls within tier threshold d if $\delta_i \leq d$. Under a uniform longitude model, the
 171 expected fraction of longitudes satisfying that inequality is $p_0 = 2d/T$. In clustered cor-
 172 pora this remains a useful geometric benchmark; the resulting p -values are interpreted
 173 as corpus-level proximity enrichment rather than counts of independent placement deci-
 174 sions. Declustering is not applied: it requires an arbitrary linkage scale with no principled
 175 stopping point, and geographic clustering here reflects documented place-specific build-
 176 ing traditions rather than duplicate observations. The multi-corpus design provides the
 177 structural safeguard: the three corpora are dominated by different geographic nodes (the
 178 distributed Eurasian corridor for UNESCO, the Java/Sumatra node for Wikidata, and
 179 crowd-sourced coverage density across South and Southeast Asia for OSM), so no single
 180 cluster can simultaneously account for enrichment across all three independently assem-
 181 bled datasets. Two empirical checks indicate the result is not a generic corridor artifact:
 182 the anchor sweep places the Levantine phase at only the 99.7th percentile of all 36,000
 183 trial longitudes (Section 5), and the UNESCO Americas sub-corpus shows no proximity
 184 enrichment at the Levantine phase (T2 rate below the geometric null).

The two null models reported in Section 5 (Null A: within-dome bootstrap; Null B: same-footprint UNESCO background, the discriminating comparison) provide complementary assessments of the observed UNESCO proximity enrichment; they are discussed in Section 6.

3.2 Tier Classification

We define three proximity thresholds as round fractions of the $T/2$ half-period (1.5°), calibrated to bracket the empirical natural proximity scales of the corpora under study: the Old World Trade Routes (OWTRAD; Ciolek 2004) Silk Road network (2.6%), the UNESCO dome subset (3.6%), the full UNESCO Cultural/Mixed corpus (10.2%), and the Wikidata stupa corpus (20.2%) (peak-enrichment null rates from the multiscale window sweep; reproduction in `supplementary/audit/tier_calibration_audit.py`). Each threshold corresponds to a geometric null rate $p_0 = 2d/T$ (T1 denotes the finest, strictest threshold; T3 the most lenient):

Tier	Threshold	Geometric null p_0
T1	$\delta \leq 0.06^\circ$ ($\lesssim 6.66$ km equatorial)	4%
T2	$\delta \leq 0.15^\circ$ ($\lesssim 16.6$ km)	10%
T3	$\delta \leq 0.3^\circ$ ($\lesssim 33$ km)	20%

Tier labels are descriptive shorthand; the primary test is always the binomial. Because the thresholds are calibrated to the empirical proximity scales of the corpora under study, data-adaptive threshold selection of this kind may favor detecting enrichment at the chosen values; results should be evaluated by the binomial statistics rather than the tier labels themselves.

3.3 Cross-Corpus Consistency via Fisher Combination

For each corpus c at tier d , a one-sided binomial test yields p_c . Fisher’s combination statistic $\chi^2 = -2 \sum_c \ln p_c$, referred to $\chi^2(2C)$ for C corpora, summarizes whether the directional excess is consistent across all corpora simultaneously. Fisher’s method formally

assumes independent p -values. The three corpora were assembled by distinct processes and their geographic clusters are substantially non-overlapping, but they share historical and geographic structure (Eurasian corridor, stupa tradition, public data sources). The combined p -value is therefore used as a descriptive consistency summary only; the individual corpus p -values are the primary evidence, and the combination summarizes their direction rather than providing a formal joint test with frequentist error guarantees.

The dome corpus period sweep identifies 3° as the dominant proximity peak in that corpus; this motivates evaluating the independently assembled corpora at the same period. Wikidata and OSM were evaluated at 3° without any period sweep of those corpora. This asymmetry is verifiable from the published code and audit files. Following [Silva \(2020\)](#), all results are treated as statistical description of spatial structure; there is no prior documentation of a 3° harmonic grid anchored to the Levantine corridor, and these statistics do not constitute evidence for one absent independent archaeological support. The two-corpus Fisher restricted to Wikidata and OSM (reported in Table 2 as “Dome-excl. combined”) summarizes whether the same period shows consistent proximity enrichment across independently assembled datasets.

4 Data

UNESCO dome/stupa subset ($N = 90$). The UNESCO World Heritage XML dataset ([UNESCO World Heritage Centre, 2024](#)) contains 1011 Cultural and Mixed inscribed sites. Seven morphological keywords (stupa/stupas, tholos, dome/domed/domes, spherical) were matched against site names, XML descriptions, and OUV text, yielding $N = 90$ sites. All statistical tests use this keyword-matched population.

Wikidata Q180987 stupa corpus ($N = 229$). All *instance-of: stupa* (Q180987) items in the Wikidata knowledge base, extracted via SPARQL query on the *instance-of: stupa* property. The corpus is geographically concentrated in the Java/Sumatra node, roughly from $\sim 105^\circ\text{E}$ to $\sim 115^\circ\text{E}$ ([Miksic, 1990](#)).

OSM stupa layer ($N = 259$). OpenStreetMap features with tags `building=stupa`,

historic=stupa, or ruins=stupa, assembled by crowd-sourced tagging via Overpass API (OpenStreetMap contributors, 2024). Data © OpenStreetMap contributors, available under the Open Database Licence (ODbL 1.0; <https://opendatacommons.org/licenses/odbl/>). The corpus is treated as a co-primary dataset: it is assembled by a process entirely independent of UNESCO and Wikidata curation, and OSM’s coverage notes are a property of the dataset as a whole rather than a limitation specific to this application.

All three corpora were tested against the same fixed candidate frame: period 3° and the ~35.25°E Levantine corridor phase. Because open datasets are publicly available, the frame is treated uniformly as fixed for the present analysis across all three corpora. The empirical application therefore functions as a frame-specified spatial test.

Candidate frame. The candidate reference phase is the ~35.25°E Levantine corridor phase, which emerged as the dominant proximity peak phase in the dome corpus sweep described in Section 2; Wikidata and OSM were evaluated at this phase without independent sweeps of their own. The counts are invariant within the band. Any anchor longitude within $\pm 0.06^\circ$ of this phase, including Jerusalem at 35.232°E, produces identical proximity counts, and the data cannot distinguish between longitudes on the same grid within the same proximity tier. The anchor is labeled by Mount Gerizim at 35.269°E, a topographic summit within the Levantine corridor absent from all monument corpora and not derived from any external literature on longitude grids. It serves as a named geographic coordinate for the interval under test; no substantive weight should be attached to the choice of Gerizim over any other co-grid longitude within the same proximity tier.

5 Results

5.1 Primary Result: Cross-Corpus Geometric-Null Test

Table 2 reports the primary statistical results: one-sided binomial tests and Fisher combinations across all three corpora.

Table 2: Cross-corpus geometric-null harmonic-proximity enrichment at the fixed 3° period and ~35.25°E Levantine corridor phase. For each corpus and tier, the one-sided binomial test compares the observed number of sites within distance d of any harmonic with the expected count under the geometric benchmark $p_0 = 2d/T$. The “Dome-excl. combined” row restricts Fisher’s combination to Wikidata and OSM (excluding the period-motivating dome corpus) and is the primary cross-corpus consistency summary; see Section 3 for the independence caveat.

Tier	Corpus	N	Hits	Exp	Fold	p_{binom}	Sig
T1 ($p_0=4\%$)	UNESCO domes	90	11	3.6	3.06×	0.0009	***
	Wikidata stupas	229	11	9.2	1.2×	0.311	ns
	OSM stupas	259	12	10.4	1.16×	0.3433	ns
	<i>All-3 Fisher</i> $\chi^2(6) = 18.402$, <i>nominal</i> $p = 0.0053$						
	<i>Dome-excl. combined (Wiki+OSM)</i> $\chi^2(4) = 4.474$, $p = 0.346$, <i>ns</i>						
T2 ($p_0=10\%$)	UNESCO domes	90	18	9.0	2.0×	0.0032	**
	Wikidata stupas	229	33	22.9	1.44×	0.0212	*
	OSM stupas	259	33	25.9	1.27×	0.089	
	<i>All-3 Fisher</i> $\chi^2(6) = 24.017$, <i>nominal</i> $p = <0.001$						
	<i>Dome-excl. combined (Wiki+OSM)</i> $\chi^2(4) = 12.547$, $p = 0.014$, *						
T3 ($p_0=20\%$)	UNESCO domes	90	29	18.0	1.61×	0.0043	**
	Wikidata stupas	229	70	45.8	1.53×	0.0001	***
	OSM stupas	259	69	51.8	1.33×	0.0059	**
	<i>All-3 Fisher</i> $\chi^2(6) = 39.661$, <i>nominal</i> $p < 0.001$						
	<i>Dome-excl. combined (Wiki+OSM)</i> $\chi^2(4) = 28.74$, $p < 0.001$, ***						

In the worked example, all three corpora exceed the geometric null at every tier despite substantially different geographic footprints and assembly processes. Fisher combinations are reported as cross-corpus consistency summaries (see Section 3 for the independence caveat). The UNESCO T1 result shows the largest individual fold-enrichment (3.06×); Wikidata and OSM T1 results are individually non-significant. The two-corpus combination restricted to Wikidata and OSM (“Dome-excl. combined”) yields $\chi^2(4) = 28.74$, $p < 0.001$ at T3 and $\chi^2(4) = 12.547$, $p = 0.014$ at T2; the T1 two-corpus combination is non-significant ($p = 0.346$). The all-three-corpus T3 Fisher combination ($p < 0.001$) includes the dome corpus, which motivated the period, and provides a less conservative cross-corpus summary than the dome-excluded combination. All p -values are frame-conditional proximity-enrichment measures for the fixed 3° period and ~35.25°E Levantine corridor phase; following Silva (2020) they are statistical descriptions of spatial structure, not evidence of historical placement. The 3° period emerged as the dominant proximity peak in the dome corpus sweep (Section 2); the three-corpus analysis is

the methodological demonstration that the same statistical pattern appears consistently across independently assembled datasets at this period.

5.2 Supplementary Rayleigh Phase Concentration

Rayleigh statistics are retained as complementary diagnostics for comparability with prior literature. Table 3 reports them at $T = 3^\circ$. The full UNESCO corpus shows no Rayleigh phase concentration ($R = 0.0224$, perm- $p = 0.6008$, ns) at $T = 3^\circ$, consistent with the proximity pattern not being a generic circular-statistical property of the full heritage list. Watson’s U^2 and Kuiper tests are retained as corroborating two-sample diagnostics (not independent tests): both show that the dome/stupa phase distribution differs detectably from the full UNESCO corpus ($p = 0.0036$, $p = 0.0013$ respectively). Because any geographically concentrated subset will tend to differ from its parent corpus in phase distribution, these results show concentration rather than proximity enrichment per se; they are included for comparability with prior literature that uses two-sample circular diagnostics as subset-specificity checks.

Table 3: Supplementary Rayleigh phase concentration at $T = 3^\circ$, retained for comparability with prior literature. The binomial proximity result (Table 2) is the matched test; Rayleigh statistics are reported here as supplementary diagnostics. All results are conditional on the fixed corridor-phase frame.

Corpus	N	R	perm- p	Sig
UNESCO full corpus	1011	0.0224	0.6008	ns
UNESCO dome/stupa	90	0.1365	0.1883	ns
Wikidata Q180987	229	0.1669	0.0016	**
OSM stupa layer	259	0.1073	0.0355	*
<i>Fisher (Wikidata + OSM): $\chi^2(4) = 19.551$, $p = 0.0006$</i>				

5.3 Route-Network Contrast, Negative Controls, and Scope

OWTRAD Silk Road route-network contrast. The OWTRAD Silk Road route network (1674 nodes, 1946 route segments) is included as an exploratory contrast. It is not a negative control: the Silk Road network occupies the same geographic range as the monument corpora, and proximity enrichment cannot be assumed absent without testing.

Two binomial proximity tests are distinguished by their sampling unit. The per-segment midpoint test uses one longitude per route segment (non-deduplicated), weighting each segment equally regardless of how many edges meet at a hub; this test is non-significant at both tiers (T1: $p = 0.0908$, ns; T2: $p = 0.0679$, ns). The degree-weighted node test weights each node by its degree centrality (the number of incident route edges), as a proxy for hub importance; this test is significant at both tiers (T1: $p = < 0.0001$; T2: $p = 0.0002$). The degree-weighted result is itself an unexplained positive finding: Eurasia spans roughly 180 of longitude and the result cannot be explained simply by noting that both trade hubs and monuments are located there; the question is why both concentrate near the same specific 3 harmonic within that space. This result therefore does not function as a negative control and is not explained by any mechanism available in these data; it is discussed further in Section 6. The NPC directional test does not independently recover the monument phase ($K = 0.7429$, perm- $p = 0.46532$, ns), indicating that the route-segment phase distribution as a whole does not share the monument mean phase direction.

The UNESCO Americas sub-corpus ($N = 145$) shows T2 rate of 9.7%, below the 10% geometric null, indicating no proximity enrichment at the $\sim 35.25^\circ\text{E}$ Levantine corridor phase in the Western Hemisphere.

A type-matched geographic control uses the North American mound corpus ($N = 236$): National Historic Landmark (NHL) and National Register of Historic Places (NRHP)-registered earthen burial and platform mounds assembled from the NPS National Register database with coordinates resolved via Wikidata ([NPS, 2024](#); [Wikimedia Foundation, 2024](#)). These are broadly mounded earthen structures used here as a geographically distinct type control, located entirely in the Western Hemisphere outside any connection to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. At the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase the mounds corpus shows no proximity enrichment at any tier: T1: 1/236 (0.4%, null 4%, $p = 1.00$, ns); T2: 11/236 (4.7%, null 10%, $p = 1.00$, ns); T3: 45/236 (19.1%, null 20%, $p = 0.66$, ns). All three tiers are at or below the geometric null. This supports the methodological illustration that the Eurasian

pattern is geographically specific to the Eurasian corridor rather than a generic outcome of applying the same harmonic-proximity test to hemispherical monument corpora.

Null B (same-footprint restricted draw, 100,000 trials) is the discriminating comparison: does the dome/stupa subset show greater harmonic proximity than the broader UNESCO background occupying the same longitude footprint? This comparison is possible for the UNESCO corpus because a larger parent population (the full UNESCO Cultural/Mixed list, $N = 1011$ sites) exists from which the dome/stupa subset was selected. No equivalent parent population exists for the Wikidata or OSM corpora, which are already operationally defined as all stupa-tagged items; a same-footprint Null B comparison therefore applies only to the UNESCO corpus. Three window widths were tested ($\pm 2^\circ$, $\pm 5^\circ$, $\pm 10^\circ$; Bonferroni $\alpha_{\text{adj}} = 0.0167$): the $\pm 2^\circ$ window gives $p = 0.0180$; the $\pm 5^\circ$ window gives $p = 0.0129$; the $\pm 10^\circ$ window gives $p = 0.0139$. All three windows survive Bonferroni correction ($\alpha_{\text{adj}} = 0.0167$), illustrating subset-specific enrichment within the shared geography across all window choices. A within-dome bootstrap (Null A, 100,000 trials) resamples from the dome sites' own longitude distribution; its expected hit rate is defined by the same geography carrying the proximity enrichment, so it cannot serve as a discriminating comparison. The observed T1 count is non-significant under this null ($p = 0.5493$, ns), consistent with the enrichment being localized to dome-site geography.

5.4 Anchor Concordance

All three corpora recover the same $\sim 35.25^\circ\text{E}$ Levantine corridor phase. The Normalized Phasor Consensus statistic K is defined as $K = |\bar{u}_A \cdot \bar{u}_B|$, where $\bar{u}_c = (\sum_k e^{i\theta_k^{(c)}}) / |\sum_k e^{i\theta_k^{(c)}}|$ (unit complex numbers as in Section 2) is the normalized mean phasor of corpus c at the candidate period, and $|\cdot|$ denotes the modulus of the dot product of two unit complex numbers. The dot-product formulation is preferred over the circular correlation coefficient (Mardia & Jupp, 2000, §8.3) because it compares phase *directions* only, stripping out amplitude information that is irrelevant when corpora differ substantially in size and clustering: $K = 1$ when the two corpora point to the same phase, $K = 0$ when they are orthogonal. Its null distribution is estimated by permutation throughout. In the worked

example, $K \geq 0.9675$ for all pairings of the three monument corpora (Table 4). The OSM + Wikidata pair ($K = 0.9695$, perm- $p = 0.15884$, ns) illustrates the power limitation of two-corpus tests when both corpora share a Southeast Asian geographic footprint: the high K indicates strong phase agreement, but the permutation null cannot be rejected with two corpora alone. The three-corpus combination reaches perm- $p = 0.02437$ with consensus anchor at 0.029° from 35°E . The primary non-nested two-corpus pair (Dome + Wikidata) is individually non-significant, reflecting the limited power of a two-corpus permutation test: high K indicates strong phase agreement, but two corpora alone cannot reject the null that such agreement arises by chance. Significance emerges when the independently assembled OSM corpus is added as a third data source, because three corpora independently converging on the same phase is rare under the permutation null even when individual pairs are underpowered. Negative controls show K drops sharply when the monument corpora are replaced with geographic or route-network controls.

Table 4: Selected NPC anchor-concordance configurations at $T = 3^\circ$ (100,000 permutation draws). The “Conservative” row pairs the broadest possible corpus (full UNESCO Cultural/Mixed) with the Wikidata stupa corpus, maximizing denominator size at the cost of corpus specificity. Negative controls (the final three rows) indicate K is not a generic Eurasian property.

Role	Corpora	K	perm- p	Note
Non-nested pair	Dome + Wikidata Stupa	0.9675	0.16287	Primary non-nested
Non-nested pair	OSM + Wikidata Stupa	0.9695	0.15884	SE-Asia geography; underpowered
Conservative	UNESCO full + Stupa	0.982	0.12243	Broadest corpus pair
Three-corpus	Dome + Stupa + OSM	0.972	0.02437	All 3 monument corpora
Neg. control	Americas + Wikidata Stupa	0.6723	0.29819	Geographic control
Contrast corpus	UNESCO + OWTRAD	0.7429	0.46532	Route network
Type control	N. American mounds ($N=236$)	n/a	n/a	No proximity enrichment

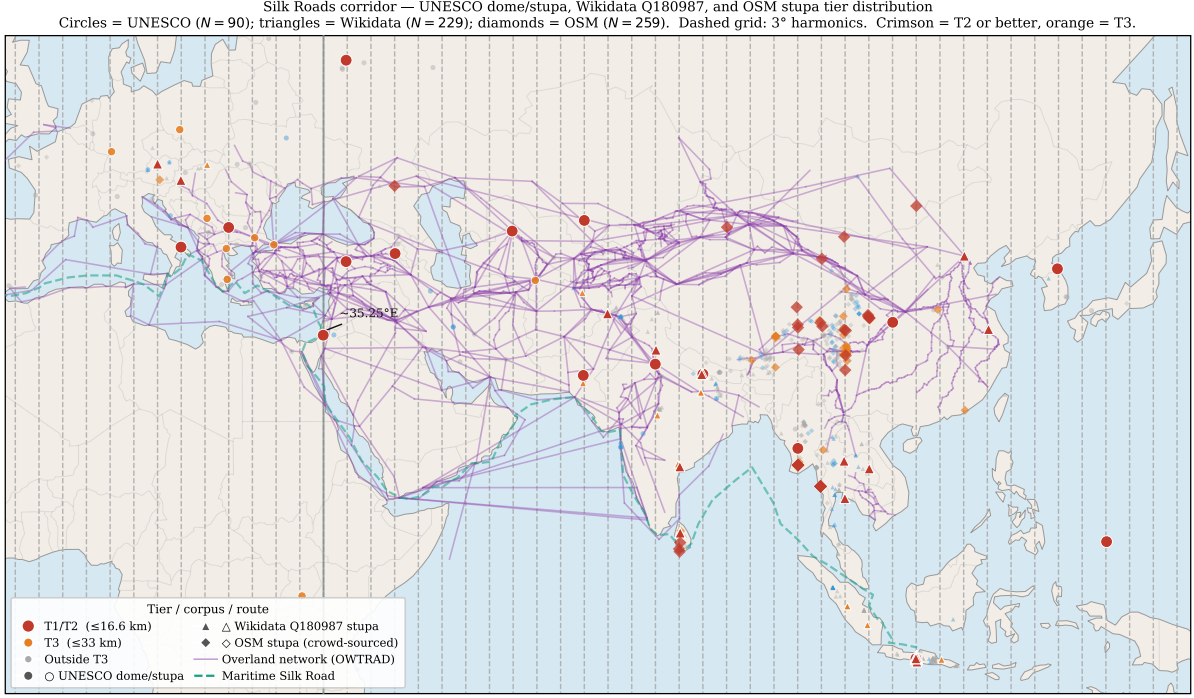


Figure 2: Silk Roads corridor showing all three monument corpora mapped against the 3° harmonic grid (dashed lines) anchored to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Circles = UNESCO dome/stupa ($N = 90$, keyword-matched); triangles = Wikidata Q180987 stupa ($N = 229$); diamonds = OSM stupa ($N = 259$, crowd-sourced). Color indicates harmonic proximity tier: crimson = T2 or better (≤ 16.6 km), orange = T3 (≤ 33 km), gray = outside T3. The dark meridian marks the fixed $\sim 35.25^\circ\text{E}$ corridor phase. Teal dashed lines show the maritime Silk Road route layer, digitised from harbour attestations in $?$, $?$, and $?$. All results are conditional on the fixed corridor-phase frame. Map projection: Mercator (WGS 84). Base map: Natural Earth (public domain, <https://www.naturalearthdata.com/>, accessed 1 March 2025).

6 Discussion

6.1 The Methods Contribution

The central methodological result is the separation between circular phase concentration and thresholded harmonic proximity. Rayleigh R describes whether the full phase distribution has a dominant mean direction at period T ; the geometric-null binomial describes whether a corpus contains more sites within a stated operational distance of any harmonic than expected from the fraction of longitude covered by those bands. Because the two statistics weight off-harmonic observations differently, they provide complementary information and can dissociate within the same corpus. This makes Rayleigh a useful diagnostic for phase structure and the geometric-null binomial the matched statistic for

thresholded proximity-to-harmonics claims. In Silva’s terms, the tests detect and describe structured features in the spatial data; archaeological, historical, or geographic explanations for those features remain interpretive questions to be evaluated with independent evidence.

The analysis is a descriptive spatial test of harmonic proximity. Geographic clustering is part of the documented corpus pattern, so site-level binomial p -values describe corpus-level proximity enrichment rather than independent cultural decisions. The geometric null $p_0 = 2d/T$ gives a transparent expected hit fraction under uniform longitude, and Fisher’s method is used as a cross-corpus consistency summary. This framework is applicable to any spatial periodicity hypothesis of this form regardless of geographic or cultural context.

6.2 Interpretation of the Empirical Illustration

The three-corpus worked example finds consistent directional excess over the geometric null at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. The dome-excluded two-corpus combination (Wikidata + OSM) yields a Fisher combination of $\chi^2(4) = 28.74$, $p < 0.001$ at T3; the all-three-corpus result (Fisher $p < 0.001$ at T3) includes the dome corpus and provides a less conservative cross-corpus summary than the dome-excluded combination. All results are frame-conditional proximity-enrichment measures (see Limitations). The UNESCO corpus is the most geographically distributed result, with T3 sites spanning the full Eurasian corridor.

Null B (same-footprint restricted draw) provides the discriminating comparison: among all UNESCO cultural sites occupying the same longitude footprint, the dome/stupa subset shows greater harmonic proximity than the broader background across all three window widths after Bonferroni correction (see Results), illustrating subset-specific enrichment within the shared geography. The cross-corpus phase concordance (NPC $K \geq 0.9675$ across all pairings of three independently assembled monument datasets with different geographic centers of mass) is consistent with the phase not being an artifact of any single corpus’s geographic distribution: UNESCO dome sites, Wikidata Q180987 stupas, and OSM stupa tags converge on the same $\sim 35.25^\circ\text{E}$ phase

independently of one another.

Corpus overlap and curation. The three corpora are assembled by distinct processes: the UNESCO dome/stupa subset derives from keyword matching against the WHC XML dataset; Wikidata Q180987 was extracted via SPARQL query on the *instance-of: stupa* property; OSM was assembled by crowd-sourced tagging of `building=stupa`, `historic=stupa`, and `ruins=stupa` features via Overpass API. An empirical overlap check using GPS proximity (threshold < 1 km) and name-string matching finds 4 near-coincident site pairs between UNESCO and Wikidata, 0 between UNESCO and OSM, and 1 between Wikidata and OSM (see supplementary overlap audit). Because all three datasets are publicly available, the candidate frame is handled as fixed across the empirical demonstration. The Fisher combination is used as a consistency check across differently assembled datasets. The geographic clusters of the three corpora are substantially non-overlapping at the region level: UNESCO dome sites span the full Eurasian corridor from the Levant to Java; Wikidata is dominated by the Java/Sumatra node; OSM coverage reflects crowd-sourced mapping density across South and Southeast Asia. No single shared geographic cluster explains enrichment in all three corpora simultaneously.

The Wikidata result is dominated by the Java/Borobudur geographic cluster (Miksic, 1990): that cluster’s position on a 3° harmonic is precisely the proximity structure the geometric-null test is designed to detect, and the geographic concentration is part of the documented corpus pattern rather than a confound to be removed. Declustering is not applied (see Section 3). The relevant check is cross-corpus: the corpus overlap audit finds only one near-coincident pair between Wikidata and OSM, so the OSM enrichment is not a restatement of the same Borobudur sites under a different label. The OSM result, assembled by a process with no relationship to UNESCO or Wikidata curation, shows the same directional excess ($p = 0.0059$ at T3) and the same phase concordance with Wikidata ($K = 0.9695$) in the worked example. Figure 2 shows all three corpora mapped against the 3° harmonic grid.

The interpretation follows Silva (2020)’s distinction between statistical structure and

archaeological explanation. Significance tests evaluate evidence against a null model; archaeological explanation is developed from evidence independent of the orientations themselves, including the wider archaeological record and socio-historical context. The statistical pattern reported here is a spatial description of three independently assembled corpora; no mechanism or historical explanation is offered. An additional positive finding complicates interpretation: weighting OWTRAD nodes by degree centrality (the number of route edges incident to each node, as a proxy for hub importance) finds that the most network-central Silk Road cities also fall near the 3° harmonic grid more often than the uniform null predicts. This is an unexplained result in its own right. It cannot be set aside by noting that both monuments and trade hubs are located in Eurasia: Eurasia spans roughly 180° of longitude, and the finding concerns proximity to a *specific* 3° harmonic within that space. Whether the monument result and the OWTRAD result are statistically independent or share some common structure cannot be determined from these data; this remains a target for future work. Historical interpretations involving deliberate placement would additionally require evidence for pre-modern continental longitude determination at sub-degree precision (Hunger & Steele, 2019; Neugebauer, 1975). The contribution of the present work is to establish what the directly matched statistical test reports when applied to three differently assembled corpora.

6.3 Limitations

Frame conditionality. All p -values in Table 2 are conditional on the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase (anchored to Mount Gerizim, 35.269°E). All three corpora are evaluated against this same frame. The results are therefore corpus-level proximity-enrichment measures under that fixed frame and should not be read as outcomes of an unconditional search over all possible periods and anchors.

Analysis sequence and exploratory scope. This paper is an exploratory methods contribution following Silva (2020). Pre-registration is a standard for confirmatory hypothesis tests; no registry exists for, and none is expected of, open-data descriptive spatial analyses of this kind. The appropriate sequence for exploratory work is to iden-

tify a statistical pattern in one dataset and assess whether it appears consistently across independently assembled datasets — which is what the dome-corpus sweep followed by the Wikidata+OSM cross-check does. The Wikidata+OSM Fisher combination is the primary cross-corpus summary precisely because it excludes the period-motivating corpus; the three-corpus result is a less conservative supplement. The asymmetry is fully disclosed and verifiable from the published code.

Anchor-sweep correction. The frame-conditional tests (Table 2) ask whether proximity enrichment is present across corpora given the fixed frame. As a separate diagnostic on the UNESCO corpus, a max-statistic permutation test over 36,000 trial longitudes asks whether the Levantine corridor phase is an unusually exceptional anchor within that corpus after accounting for the anchor search. The corrected $p = 0.0753$ ($z = 1.65$, marginal) indicates it is not strongly exceptional: the Levantine phase falls near the center of the null distribution of best-anchor counts from uniformly random data. This test was computed on the UNESCO corpus only; no anchor sweep was conducted on Wikidata or OSM, so no equivalent correction applies to those corpora. The Wikidata and OSM results are simply frame-conditional on the anchor identified in the dome sweep. The frame-conditional p -values quantify enrichment given the fixed frame; they do not quantify how exceptional that frame is within the UNESCO corpus, which is what the anchor-sweep diagnostic addresses.

7 Conclusion

This paper demonstrates the distinction between Rayleigh concentration and thresholded harmonic proximity as descriptors of a modulo- T phase distribution: Rayleigh summarizes mean circular concentration, while the geometric-null binomial tests thresholded proximity to any harmonic under the benchmark $p_0 = 2d/T$. The period sweep illustrates this distinction empirically, with Rayleigh-significant periods lacking proximity enrichment and the enriched 3.0° period lacking significant Rayleigh concentration. Applied under a fixed operational frame, with fixed period, corridor phase, thresholds, and

corpora, the binomial framework finds a consistent corpus-level proximity pattern across three hemispherical-monument datasets at the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase.

Two qualifications contextualize these results, both detailed in the Limitations section. The period emerged from the dome corpus sweep; the Wikidata+OSM combination ($\chi^2(4) = 28.74$, $p < 0.001$ at T3) is the primary cross-corpus check precisely because it excludes that corpus. A separate anchor-sweep diagnostic on the UNESCO corpus ($p = 0.0753$) shows that the Levantine corridor phase is not strongly exceptional within that corpus once the 36,000-anchor search is accounted for; this test applies only to the UNESCO corpus and does not modify the frame-conditional Wikidata+OSM result.

Following Silva (2020), the worked example is best read as a descriptive spatial exercise. This paper is a methods contribution: it introduces and validates the geometric-null binomial framework, demonstrates that Rayleigh concentration and geometric-null binomial proximity can dissociate within the same corpus while each statistic remains useful for its own question, and uses three hemispherical-monument corpora as a worked example to illustrate how the method behaves on real data. No confirmatory historical judgement is made, and none is possible from these results alone. The statistics report corpus-level proximity enrichment under a fixed frame; they cannot identify causal or intentional relationships. This is an open-data observational analysis: the datasets are publicly available and every step is independently reproducible from the published code and audit files. The evidentiary gap between “sites are disproportionately close to a 3° harmonic grid” and “sites were placed with knowledge of that grid” lies outside the scope of any purely statistical test; such claims require independent archaeological and socio-historical evidence that proximity statistics cannot supply. It also provides a reproducible open-data example that future comparative work can evaluate.

All data, code, and audit files are openly available at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.20123709>).

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Data Availability

All data, code, and audit files: Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.20123709>, confirmed resolving May 2026). UNESCO XML (UNESCO World Heritage Centre, 2024).

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CRediT Author Statement

Seth Tenenbaum: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing: original draft, Writing: review & editing, Visualization.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the author used Claude (Anthropic) to assist with prose drafting and GitHub Copilot (Microsoft) to assist with code autocompletion. After using these tools, the author reviewed and edited all content as necessary and takes full responsibility for the content of the published article.